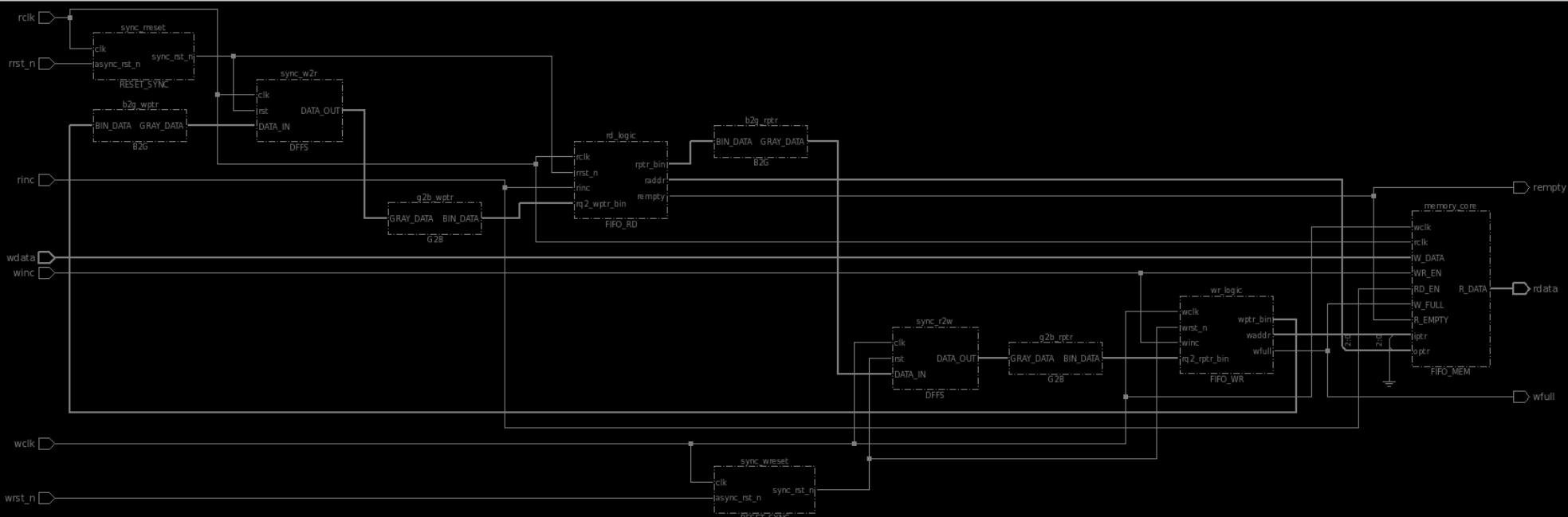
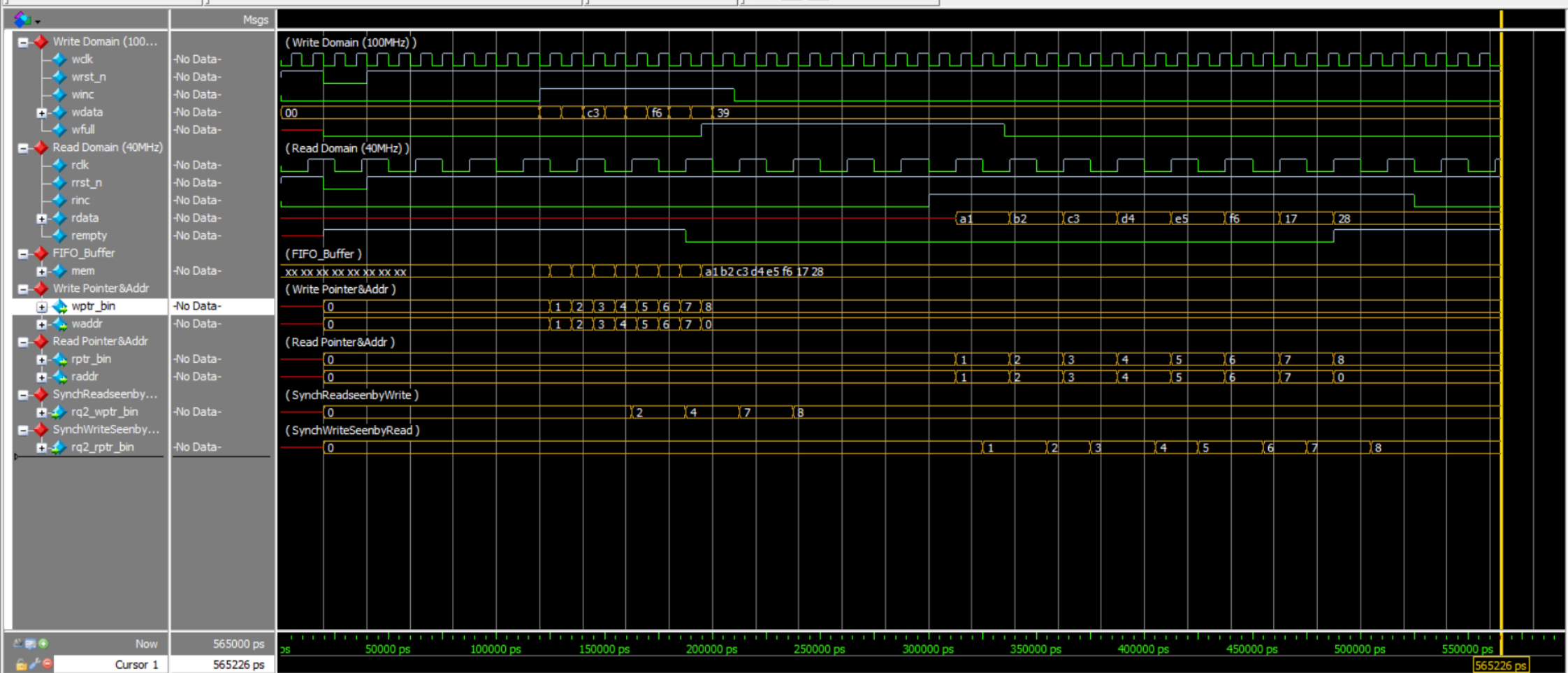

rclk
rrst_n
sync_rreset
clk
sync_rst_nj
async_rst_n
RESET_SYNC
sync_w2r
clk
rst
DATA_OUT
DATA_IN
DFFS
b2g_wptr
BIN_DATA
GRAY_DATA
B2G
rinc
g2b_wptr
GRAY_DATA
BIN_DATA
G2B
rd_logic
rclk
rrst_n
rinc
rq2_wptr_bin
rptr_bin
raddr
rempty
FIFO_RD
b2g_rptr
BIN_DATA
GRAY_DATA
B2G
rempty
wdata
winc
memory_core
wclk
rclk
W_DATA
WR_EN
RD_EN
R_DATA
W_FULL
R_EMPTY
iptr
optr
FIFO_MEM
rdata
sync_r2w
clk
rst
DATA_OUT
DATA_IN
DFFS
g2b_rptr
GRAY_DATA
BIN_DATA
G2B
wr_logic
wclk
wrst_n
winc
rq2_rptr_bin
wptr_bin
waddr
wfull
FIFO_WR
wfull
wclk
wrst_n
sync_wreset
clk
sync_rst_nj
async_rst_n
RESET_SYNC


Msgs
Write Domain (100...
wclk
wrst_n
winc
wdata
wfull
Read Domain (40MHz)
rclk
rrst_n
rinc
rdata
rempty
FIFO_Buffer
mem
Write Pointer&Addr
wptr_bin
waddr
Read Pointer&Addr
rptr_bin
raddr
SynchReadseenby...
rq2_wptr_bin
SynchWriteSeenby...
rq2_rptr_bin
-No Data-
(Write Domain (100MHz))
(Read Domain (40MHz))
(FIFO_Buffer)
(Write Pointer&Addr)
(Read Pointer&Addr)
(SynchReadseenbyWrite)
(SynchWriteSeenbyRead)
00 c3 f6 39
a1 b2 c3 d4 e5 f6 17 28
xx xx xx xx xx xx xx xx
Now
565000 ps
Cursor 1
565226 ps
50000 ps
100000 ps
150000 ps
200000 ps
250000 ps
300000 ps
350000 ps
400000 ps
450000 ps
500000 ps
550000 ps

```
VSIM(paused)> run -all
# [PASS] Post-Reset Phase (Empty) | wfull: 0 | rempty: 1 time 115000
#
# --- Starting Burst Write of 9 Bytes (100MHz) ---
# [PASS] wfull asserted exactly after 8 writes | wfull: 1 | rempty: 0 time 200000
# [PASS] Flags Stable: FIFO Full, Not Empty | wfull: 1 | rempty: 0 time 285000
#
# --- Starting Burst Read of 9 Bytes (40MHz) ---
# [PASS] Read Byte 1 | rdata: a1 time 325000
# [PASS] Read Byte 2 | rdata: b2 time 350000
# [PASS] Read Byte 3 | rdata: c3 time 375000
# [PASS] Read Byte 4 | rdata: d4 time 400000
# [PASS] Read Byte 5 | rdata: e5 time 425000
# [PASS] Read Byte 6 | rdata: f6 time 450000
# [PASS] Read Byte 7 | rdata: 17 time 475000
# [PASS] Read Byte 8 | rdata: 28 time 500000
# [PASS] rempty asserted exactly after 8 reads | wfull: 0 | rempty: 1 time 500000
# [PASS] Flags Stable: FIFO Empty, Not Full | wfull: 0 | rempty: 1 time 565000
#
# ================================================
#    ALL FIFO TESTS COMPLETED.
# ================================================
# ** Note: $finish    : ASYNC_FIFO_TB.v(166)
#    Time: 565 ns  Iteration: 0  Instance: /ASYNC_FIFO_TB
# 1
```